

Chencheng Liang

PhD in Computer Science

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Work Experience

2025-08 – 2026-06

Independent AI and Automation Consultant

- Advised small business owners on adopting AI tools and workflow automation to improve operational efficiency and reduce manual work
- Developed tailored AI training programs for business owners, enabling practical adoption of LLMs, automation, and AI-driven workflows
- Provided market research and strategic analysis for entrepreneurs exploring AI-enabled business opportunities and digital transformation initiatives

2025-12 – 2026-04

Baobaozhu (Haikou) Cross-Border E-Commerce Co., Ltd., China

Founder & Operator

- Built and operated a Shopify-based e-commerce platform, owning the full workflow from supplier onboarding and product sourcing to fulfillment and customer support
- Developed LLM-based workflows to automate product catalog creation, including attribute extraction, content generation, categorization, search engine optimization metadata, and multilingual localization
- Automated generation of product images, marketing assets, and social media content using AI workflows, significantly reducing manual effort in content production

2025-08 – 2025-11

Hainan Jiangshuo Industrial Co., Ltd., China

Co-founder & Operator

- Built workflows combining web scraping, image-based extraction, and LLMs to convert unstructured product information into structured datasets
- Developed data pipelines to collect and standardize multilingual product data from online retailers for market and competitor analysis
- Built LLM-based workflows with structured outputs to automate product attribute extraction and reduce manual data processing

2019-01 – 2025-06

Uppsala University, Sweden

PhD Student & Teaching Assistant

- Conducted research in training Graph Neural Networks (GNNs) to improve automated reasoning systems
- Developed an novel GNN-based constraint solver framework
- Supervised multiple master's projects spanning both academic research and companies such as Ericsson Research
- Taught courses in machine learning, programming methodology, and embedded systems

Education

2019 – 2025	Uppsala University, Sweden <i>PhD in Computer Science</i> Thesis: “Learning to Guide Automated Reasoning: A GNN-Based Framework” Advisors: Philipp Rümmer, Yi Wang, Marc Brockschmidt Opponent: Stephan Schulz
2016 – 2018	University of Göttingen, Germany <i>Master of Science in Computer Science</i> Thesis: “Topology Control Using Fuzzy Game Theory in Mobile Underwater Sensor Networks” Grade: 1.0 (Sehr gut, highest distinction)
2010 – 2014	Chengdu University, China <i>Bachelor of Science in Software Engineering</i> Graduated 1st in class

Teaching Experience

2019 – 2025	Uppsala University, Sweden <i>Course Instructor & Teaching Assistant</i> • Natural Computation Methods for Machine Learning (Master’s, 10 credits, 2019-2025) • Imperative and Object-Oriented Programming Methodology (Bachelor’s, 10 credits, 2023-2024) • Project CS (Master’s project course, 30 credits, 2022-2024) • Introduction to Studies in Embedded Systems (Master’s, 5 credits, 2019-2021)
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Thesis & Project Supervision

2024	Federated Learning meets Large Language Models <i>Master’s Project Supervision</i> Students: Mahtab Mirhaj, Oguzhan Ersoy, Yixin Huang, Shiqi Shu Ericsson Research & Uppsala University
2023	Distributed Mobility Prediction in Telecommunication Networks <i>Master’s Project Supervision</i> Students: Shreyansh Singh, Lifang Zheng, Savvas Giortsis, Tobias Lass, Yilei Zheng, Salum Nassor Ericsson Research & Uppsala University
2022	Shortest Path Reliable Estimation: Applying Graph Neural Networks - SpreadNet <i>Master’s Project Supervision</i> Students: ChanVuth Chea, Boli Gao, Jennifer Gross, Haouyuan Li, Paarth Sanhotra, George-Alexandru Stoian, Sofia Afnaan Syed, Haodong Zhao Ericsson Research & Uppsala University
2021	Exploring Properties and Limitations of Graph Neural Networks (GNNs) in Software Verification <i>Master’s Thesis Supervision</i> Student: Kexin Xu Uppsala University

Notable Projects & Achievements

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| 2025 | CozySteps - LLM-powered Activity Recommender
<i>Impact Hack 2025, AI Sweden</i> <ul style="list-style-type: none">• Built mobile app prototype for families with autistic children• Implemented LLM-based activity recommendation engine• Won recognition for addressing real societal challenges |
| 2023 | Insilico - AI-Driven Art Installation
<i>Uppsala Art Museum</i> <ul style="list-style-type: none">• Integrated AI-based sentiment analysis for Mat Collishaw's art installation• Developed real-time Twitter sentiment analysis system• Connected digital hostility to physical art expression |

Research Interests & Skills

- **Primary Research Areas:** Neural network training, Graph Neural Networks (GNNs), Symbolic reasoning, AI workflow automation
- **Technical Skills:** Python, Scala, Java, C/C++ , PyTorch, TensorFlow, Git, Linux, Docker
- **AI/ML:** Deep Learning (CNN, RNN, GNN, Transformers), Reinforcement Learning, NLP, Classical ML
- **Languages:** Chinese (Native), English (Fluent), Japanese (Basic)

Professional Service

- **Subreviewer for Conferences:**
 - FMCAD 2025, FM 2024, LPAR 2024, VMCAI 2024, FMCAD 2023, LPAR 2023
 - FMCAD 2022, IJCAR 2022, FroCoS 2021, CADE 2021, TACAS 2020
 - iFM 2019, FroCoS 2019, CADE 2019
- **Talks & Presentations:**
 - “Guiding Constraint Horn Clauses Solving using Graph Neural Networks” - ISTA, Austria, 2024
 - “Boosting Constrained Horn Solving by Unsat Core Learning” - HCVS Workshop, Luxembourg, 2024
 - “Exploring Representation of Horn Clauses using GNNs” - AITP Conference, France, 2022
 - “Guiding Interpolation for Model Checking by Deep Learning Techniques” - AVM, Czech Republic, 2019

Publications

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| 2025 | 1. | Abdulla, P. A., Atig, M. F., Cailler, J., Liang, C. & Rümmer, P. <i>When GNNs Met a Word Equations Solver: Learning to Rank Equations in 15th International Symposium on Frontiers of Combining Systems (FroCoS)</i> (Springer, 2025). |
| 2024 | 2. | Abdulla, P. A., Atig, M. F., Cailler, J., Liang, C. & Rümmer, P. <i>Guiding Word Equation Solving Using Graph Neural Networks in 22nd International Symposium on Automated Technology for Verification and Analysis (ATVA)</i> (Springer, 2024). |
| | 3. | Abdulla, P. A., Liang, C. & Rümmer, P. <i>Boosting Constrained Horn Solving by Unsat Core Learning in 25th International Conference on Verification, Model Checking, and Abstract Interpretation (VMCAI)</i> (Springer, 2024). |
| 2022 | 4. | Liang, C. , Rümmer, P. & Brockschmidt, M. <i>Exploring Representation of Horn clauses using GNNs in 8th Workshop on Practical Aspects of Automated Reasoning (PAAR)</i> (2022). |
| 2021 | 5. | Yuan, Y., Liang, C. , Chen, X., Baker, T. & Fu, X. Adaptive Fuzzy Game-Based Energy-Efficient Localization in 3D Underwater Sensor Networks. <i>ACM Transactions on Internet Technology</i> 22 , 1–20 (2021). |
| 2019 | 6. | Yuan, Y., Liang, C. , Kaneko, M., Chen, X. & Hogrefe, D. Topology Control for Energy-Efficient Localization in Mobile Underwater Sensor Networks Using Stackelberg Game. <i>IEEE Transactions on Vehicular Technology</i> 68 , 1487–1500 (2019). |
| 2017 | 7. | Yang, Y., Liang, C. & Ji, S. Comments on “Fuzzy multicriteria decision making method based on the improved accuracy function for interval-valued intuitionistic fuzzy sets” by Ridvan Sahin. <i>Soft Computing</i> 21 , 3033–3035 (2017). |
| 2015 | 8. | Yang, Y., Liang, C. , Ji, S. & Liu, T. Adjustable soft discernibility matrix based on picture fuzzy soft sets and its applications in decision making. <i>Journal of Intelligent & Fuzzy Systems: Applications in Engineering and Technology</i> 29 , 1711–1722 (2015). |